

Stat 5200, Module 4: Factorial Design and Analysis

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1 Overview

Objectives and overview

- This material is based on Chapter 8 of the textbook “A First Course in Design and Analysis of Experiments” (Oehlert, 2000).
- Module outcomes include:
 - You will understand complete and incomplete factorial treatment structures.
 - You will develop an intuitive and formal understanding of interaction effects.
 - You will be able to specify and fit additive and non-additive two-way ANOVA models.
 - You will learn how to construct and interpret main effect contrasts and interaction contrasts.
 - You will know how to fit and analyze factorial designs in R using `aov()` and `emmeans()`.

2 Factorial Treatment Structure

Introduction to Factorial Structures

- In many experiments, we want to evaluate the effect of two or more explanatory factors simultaneously.

	Beef ($j = 1$)	Cereal ($j = 2$)	Pork ($j = 3$)	Marginal Row Mean
High Intake ($i = 1$)	μ_{11}	μ_{12}	μ_{13}	$\bar{\mu}_{1\bullet}$
Low Intake ($i = 2$)	μ_{21}	μ_{22}	μ_{23}	$\bar{\mu}_{2\bullet}$
Marginal Col Mean	$\bar{\mu}_{\bullet 1}$	$\bar{\mu}_{\bullet 2}$	$\bar{\mu}_{\bullet 3}$	μ (Grand Mean)

- A *factorial treatment structure* defines treatments as combinations of levels of each factor.
- Suppose Factor A has a levels and Factor B has b levels.
- A *complete factorial design* includes all $g = a \times b$ possible treatment combinations.

Example 1: 2×3 Diet Experiment

- Factor 1: Protein Source (Beef, Cereal, Pork) $\implies b = 3$.
- Factor 2: Protein Intake (High, Low) $\implies a = 2$.
- Total treatment combinations: $g = 2 \times 3 = 6$ unique diets.

Example 2: 2×2 Fertilizer Experiment

- Factor A: Superphosphate (P) $\rightarrow$ Levels: {Not Added, Added}
- Factor B: Potash (K) $\rightarrow$ Levels: {Not Added, Added}
- $g = 2 \times 2 = 4$ treatments:
 1. None (P_1K_1)
 2. Superphosphate only (P_2K_1)
 3. Potash only (P_1K_2)
 4. Both Superphosphate and Potash (P_2K_2)

Notice the power of a factorial design: instead of running two separate experiments (one for Superphosphate and one for Potash), we run a single experiment that evaluates both factors simultaneously using the exact same experimental units. This yields higher statistical efficiency and allows us to study whether the two factors work together or interfere with each other.

Main Effects and Interactions

- We are generally interested in both the *main effects* of each treatment, but also the *interactions* between treatments. Consider the example below (Example 2):
- **Main Effects:** Evaluated by comparing the marginal row means ($\bar{\mu}_{1\bullet}$ vs. $\bar{\mu}_{2\bullet}$) or the marginal column means ($\bar{\mu}_{\bullet 1}$, $\bar{\mu}_{\bullet 2}$, $\bar{\mu}_{\bullet 3}$). For example, what is the difference in response between high and low intake treatments?
- **Interactions:** Evaluated by looking at differences between cells. For example, is the difference between High and Low intake the same for Beef as it is for Cereal?

$$(\mu_{11} - \mu_{21}) \stackrel{?}{=} (\mu_{12} - \mu_{22})$$

- **The ANOVA Alternative:** While we *could* formalize and test all these specific comparisons using individual contrasts (which we discussed in previous modules), doing so for every combination is tedious and inflates our error rate.
- Instead, we will introduce the *Two-Way ANOVA* as an efficient way to test for overall main effects and interactions simultaneously.
- Like one-way ANOVA, we will express the total sum of squares in terms of treatments and errors.

If we denote A as the first treatment type, B as the second, and AB as the interaction, then we will have a similar break down:

$$SS_T = SS_A + SS_B + SS_{AB} + SS_E$$

Notation for Two-Way Factorials

- To identify observations in a two-way design, we use three subscripts:

$$y_{ijk}$$

- $i = 1, \dots, a$: level of Factor A
- $j = 1, \dots, b$: level of Factor B
- $k = 1, \dots, n$: replication number within treatment combination (i, j)

- Summary Notation:

- Sample mean for treatment (i, j) : $\bar{y}_{ij\bullet} = \frac{1}{n} \sum_{k=1}^n y_{ijk}$
- Marginal mean for level i of Factor A: $\bar{y}_{i\bullet\bullet} = \frac{1}{bn} \sum_{j=1}^b \sum_{k=1}^n y_{ijk}$
- Marginal mean for level j of Factor B: $\bar{y}_{\bullet j\bullet} = \frac{1}{an} \sum_{i=1}^a \sum_{k=1}^n y_{ijk}$
- Overall grand mean: $\bar{y}_{\bullet\bullet\bullet} = \frac{1}{abn} \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n y_{ijk}$

3 The Two-Way ANOVA Model

Model Specification

- For completely randomized designs, we will start with *balanced factorial* treatments. All $n_{ij} = n$ for all i, j .
- We model the response y_{ijk} as:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where:

- μ : Overall grand mean.
- α_i : Main effect of level i of Factor A.
- β_j : Main effect of level j of Factor B.
- $(\alpha\beta)_{ij}$: Interaction effect between level i of A and level j of B.
- $\epsilon_{ijk} \stackrel{iid}{\sim} N(0, \sigma^2)$: Random errors.

- *Parameter Identification Restrictions (Sum-to-Zero Constraints):*

$$\sum_{i=1}^a \alpha_i = \sum_{j=1}^b \beta_j = \sum_{i=1}^a (\alpha\beta)_{ik} = \sum_{j=1}^b (\alpha\beta)_{kj} = 0 \text{ for all } k$$

- *Hierarchy of Models:*

1. **Full (Non-additive) Model:** $y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$
2. **Additive Model:** $y_{ijk} = \mu + \alpha_i + \beta_j + \epsilon_{ijk}$
3. **Main Effect Models:** $y_{ijk} = \mu + \alpha_i + \epsilon_{ijk}$, $y_{ijk} = \mu + \beta_j + \epsilon_{ijk}$
4. **Constant Model:** $y_{ijk} = \mu + \epsilon_{ijk}$

Degrees of Freedom

- When computing the degrees of freedom for each parameter, note that the complete factorial structure performs and ANOVA for treatment A and B simultaneously.

The degrees of freedom for each main effect is computed as we would in a one-way ANOVA:

- Main effect *A*: $(a - 1)$ where a is the number of treatment levels for factor *A*.
- Main effect *B*: $(b - 1)$ where b is the number of treatment levels for factor *A*.
- Interaction effect (*AB*): $(a - 1)(b - 1)$.

One way to think about this last degree of freedom calculation is to see how many degrees of freedom are unused if we just treated it as a one-way ANOVA for the individual treatment groups μ_{ij} : There are ab total treatment means and we lose one degree of freedom for the grand mean, so we have $ab - 1$ total degrees of freedom that must be accounted for. If there are $(a - 1)$ for the main effects of factor *A*, and $(b - 1)$ for the main effects of factor *B*, then there must be:

$$(ab - 1) - (a - 1) - (b - 1) = ab - a - b + 1 = (a - 1)(b - 1)$$

degrees of freedom left for the interaction effects.

4 Understanding Interaction Effects

What is an Interaction Effect?

- *Definition:* An interaction effect occurs when the effect of one factor on the response depends on the level of another factor.
- Ask yourself: "*Does the difference between levels of Factor A stay the same regardless of Factor B?*"
 - If **YES**: The factors are *additive* (No Interaction).
 - If **NO**: There is an *interaction* between factors.

Intuition: "Difference of Differences"

Consider a 2×2 design with population means μ_{ij} :

- Simple main effect of A when Factor B is at level 1: $\mu_{11} - \mu_{21}$
- Simple main effect of A when Factor B is at level 2: $\mu_{12} - \mu_{22}$

The *interaction effect* is the difference between these two simple effects:

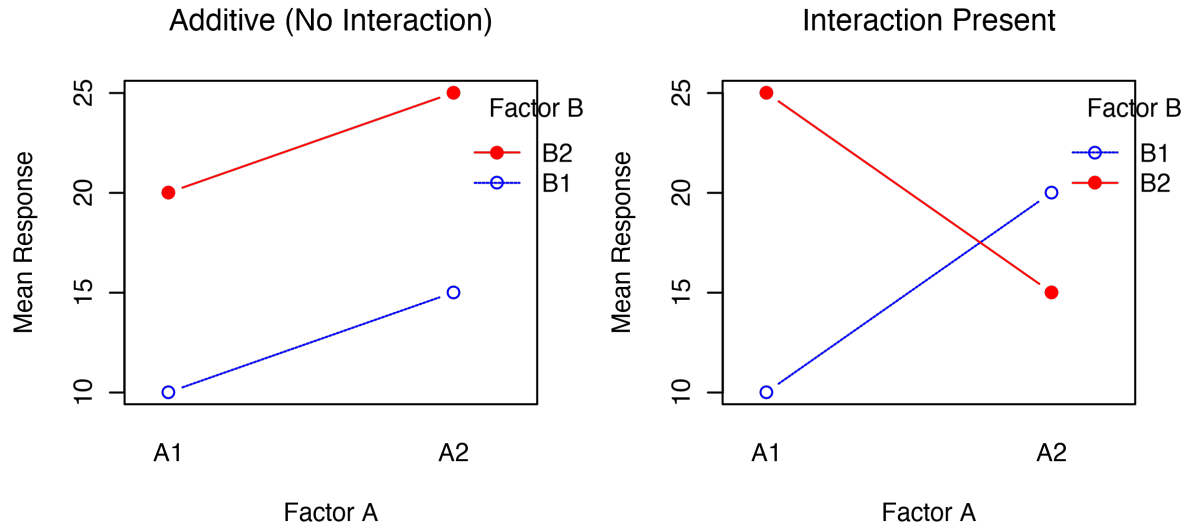
$$(\mu_{11} - \mu_{21}) - (\mu_{12} - \mu_{22})$$

If this difference equals zero, the effect of A is constant across levels of B. If non-zero, an interaction exists.

Source	DF	Sum of Squares (SS)	Mean Square	F-Statistic
Factor A	$a - 1$	$SS_A = bn \sum \hat{\alpha}_i^2$	$MS_A = \frac{SS_A}{a-1}$	$F_A = \frac{MS_A}{MS_E}$
Factor B	$b - 1$	$SS_B = an \sum \hat{\beta}_j^2$	$MS_B = \frac{SS_B}{b-1}$	$F_B = \frac{MS_B}{MS_E}$
Interaction AB	$(a - 1)(b - 1)$	$SS_{AB} = n \sum \sum \hat{\alpha}\hat{\beta}_{ij}^2$	$MS_{AB} = \frac{SS_{AB}}{(a-1)(b-1)}$	$F_{AB} = \frac{MS_{AB}}{MS_E}$
Error	$ab(n - 1)$	$SS_E = \sum \sum \sum (y_{ijk} - \bar{y}_{ij\bullet})^2$	$MS_E = \frac{SS_E}{ab(n-1)}$	
Total	$abn - 1$	$SS_T = SS_A + SS_B + SS_{AB} + SS_E$		

Graphical Diagnosis: Interaction Plots

- An *interaction plot* displays treatment group means across levels of one factor, with lines connecting levels of the second factor.
- **Parallel lines** $\implies$ No Interaction (Additive).
- **Non-parallel or crossing lines** $\implies$ Interaction present.



5 ANOVA Mechanics and Hypothesis Testing

Two-Way ANOVA Table (Balanced Design)

- For a balanced design (n observations per cell, $N = abn$ total):

Sequence of Testing: Top-Down Rule

1. **Step 1: Test Interaction** ($H_0 : (\alpha\beta)_{ij} = 0$ for all i, j)
 - If **Significant**: Stop! The model is non-additive. Do not interpret main effect F -tests in isolation. Examine simple main effects or interaction contrasts.
 - If **Not Significant**: Drop interaction term to obtain the additive model.
2. **Step 2: Test Main Effects**
 - If interaction is non-significant, proceed to test $H_0 : \alpha_i = 0$ and $H_0 : \beta_j = 0$ independently using the main effect F -tests.

6 Applied Examples in R

Example 1: Significant Interaction (ToothGrowth)

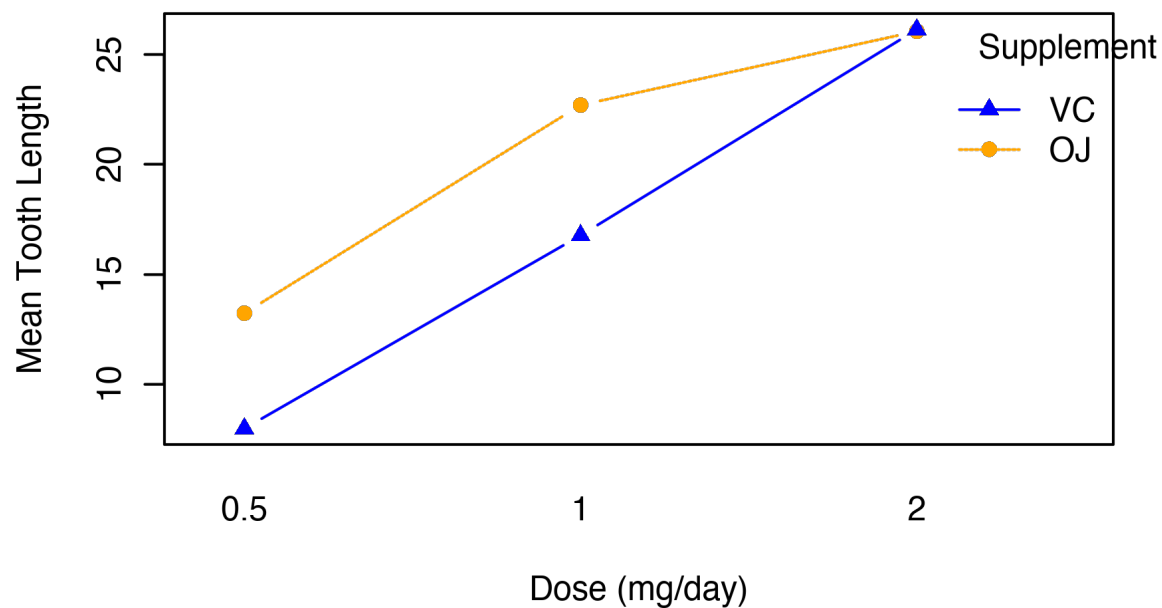
- The built-in `ToothGrowth` dataset measures tooth length (`len`) in guinea pigs under two factors:
 - **Supplement (supp):** Orange Juice (OJ) or Ascorbic Acid (VC).
 - **Dose (dose):** 0.5, 1.0, or 2.0 mg/day.
- First, we must convert `dose` from a numeric variable into a factor so R treats it as categorical.

```
# Load data and convert dose to a factor
data(ToothGrowth)
ToothGrowth$dose <- as.factor(ToothGrowth$dose)
```

1. Visualizing the Interaction

We start with an interaction plot. Crossing or converging lines suggest an interaction effect.

```
interaction.plot(
  x.factor = ToothGrowth$dose,
  trace.factor = ToothGrowth$supp,
  response = ToothGrowth$len,
  type = "b", pch = c(16, 17),
  col = c("orange", "blue"),
  xlab = "Dose (mg/day)",
  ylab = "Mean Tooth Length",
  trace.label = "Supplement"
)
```



2. Fitting the Model and Testing (Top-Down Rule)

We fit the full model with the `*` operator and check the ANOVA table.

```
mod_tooth <- aov(len ~ supp * dose, data = ToothGrowth)
summary(mod_tooth)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
supp	1	205.4	205.4	15.572	0.000231	***
dose	2	2426.4	1213.2	92.000	< 2e-16	***
supp:dose	2	108.3	54.2	4.107	0.021860	*
Residuals	54	712.1	13.2			

Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- **Interpretation:** The `supp:dose` interaction is significant ($p = 0.02$). Following our Top-Down Rule, we **stop** and do not interpret the main effects blindly. The effect of the supplement depends entirely on the dose.

3. Analyzing Simple Effects with emmeans

Because the interaction is significant, we should look at the “simple effects”—the effect of the supplement *at each specific level* of the dose.

```
library(emmeans)
# The "/" operator tells emmeans to compare 'supp' AT each level of 'dose'
emm_tooth <- emmeans(mod_tooth, pairwise ~ supp | dose)

# View the pairwise comparisons
emm_tooth$contrasts
```

dose = 0.5:

contrast	estimate	SE	df	t.ratio	p.value
OJ - VC	5.25	1.62	54	3.233	0.0021

dose = 1:

contrast	estimate	SE	df	t.ratio	p.value
OJ - VC	5.93	1.62	54	3.651	0.0006

dose = 2:

contrast	estimate	SE	df	t.ratio	p.value
OJ - VC	-0.08	1.62	54	-0.049	0.9609

- **Conclusion:** Orange juice (OJ) results in significantly longer teeth at doses of 0.5 and 1.0, but there is no significant difference between supplements at a dose of 2.0.

Demystifying the emmeans Syntax

- In a one-way ANOVA, you likely used `emmeans(model, ~ Factor)` to get the estimated means for each group.
- In a two-way ANOVA with a significant interaction, we must examine *simple main effects*. We need to know the effect of one factor *at specific levels* of the other factor.

Breaking down: `pairwise ~ supp | dose`

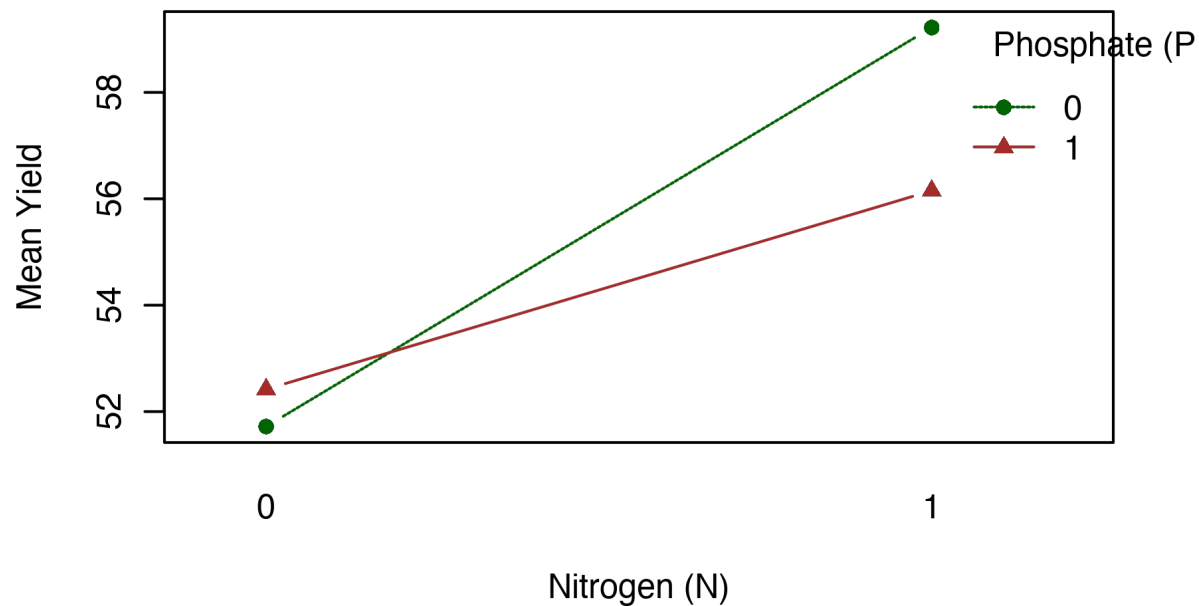
- `pairwise ~ supp`: This tells R to calculate the pairwise differences between the levels of the Supplement factor (OJ vs. VC).
- `|` (The Pipe/Bar): In R formulas, the vertical bar means "given" or "conditioned on."
- `dose`: This is the grouping variable. It tells R to split the analysis up by the levels of Dose.
- **Plain English Translation:** "Compare the mean tooth length of OJ vs. VC, and do this comparison three separate times: once for the 0.5 dose, once for the 1.0 dose, and once for the 2.0 dose."

Example 2: No Interaction (Additive Model)

- The built-in `npk` dataset records pea yields based on applying Nitrogen (N) and Phosphate (P).
- Let's evaluate if Nitrogen and Phosphate interact to affect yield.

```
data(npk)

# Visualize the data
interaction.plot(x.factor = npk$N, trace.factor = npk$P,
  response = npk$yield,
  type = "b", pch = c(16, 17), col = c("darkgreen", "brown"),
  xlab = "Nitrogen (N)", ylab = "Mean Yield",
  trace.label = "Phosphate (P)")
```



1. Fitting the Full Model

Notice in the plot that the lines are nearly perfectly parallel. We test this statistically by fitting the full factorial model.

```
mod_npk <- aov(yield ~ N * P, data = npk)
summary(mod_npk)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
N	1	189.3	189.28	5.758	0.0263 *
P	1	8.4	8.40	0.256	0.6187
N:P	1	21.3	21.28	0.647	0.4305
Residuals	20	657.4	32.87		

Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- **Interpretation:** The N:P interaction is not significant ($p = 0.485$). We can drop the interaction term and fit an *additive model*.

2. Fitting the Additive Model

We use the + operator to fit a model assuming the effects of N and P are strictly independent.

```
# Fit additive model (no interaction)
mod_npk_add <- aov(yield ~ N + P, data = npk)
summary(mod_npk_add)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
N	1	189.3	189.28	5.857	0.0247 *
P	1	8.4	8.40	0.260	0.6155
Residuals	21	678.7	32.32		

Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- **Conclusion:** Nitrogen (N) has a significant main effect on yield ($p = 0.015$), while Phosphate (P) does not ($p = 0.613$). Because the model is additive, we know Nitrogen increases yield regardless of whether Phosphate is applied or not.

7 Factorial Contrasts

Main Effects vs. Interaction Contrasts

- A contrast in a factorial design is a linear combination of treatment means $\sum_i \sum_j w_{ij} \mu_{ij}$ subject to $\sum_i \sum_j w_{ij} = 0$.
- *Main Effect Contrasts for A:* Weights for a given level of A are identical across all levels of B.

$$\sum_{i=1}^a w_i = 0, \quad w_{ij} = w_i \quad \text{for all } j$$

- *Interaction Contrasts ("Contrast of Contrasts")*: Weights are constructed as products of main effect contrast weights: $w_{ij} = w_{A,i} \times w_{B,j}$. Row sums and column sums of weights must **all equal zero**:

$$\sum_{i=1}^a w_{ij} = 0 \text{ for all } j, \quad \text{and} \quad \sum_{j=1}^b w_{ij} = 0 \text{ for all } i$$

Example: 2×2 Factorial Weight Matrices

Main Effect A (W_A)

1	1
-1	-1

Main Effect B (W_B)

1	-1
1	-1

Interaction (W_{AB})

1	-1
-1	1

8 Advanced Topics in Practice (R)

Unbalanced Data: The Order Matters

- Real-world data is rarely perfectly balanced (e.g., dropped samples, survey non-response).
- When a design is *unbalanced*, factor effects are no longer strictly independent (orthogonal).
- **The Trap**: Base R's `aov()` uses **Type I (Sequential) Sums of Squares**. The order you list your variables completely changes the p-values!

```
# In an unbalanced dataset, these two models test
# different hypotheses for the main effects!
mod1 <- aov(yield ~ P * K, data = unbalanced_data)
summary(mod1)

mod2 <- aov(yield ~ K * P, data = unbalanced_data)
summary(mod2)
```

Best Practices for Unbalanced Data

1. **Use `joint_tests()`**: Base R's contrast coding makes standard Type III ANOVA tables difficult to calculate correctly. Instead, use `joint_tests()` from the **emmeans** package to get an ANOVA-like table that is mathematically correct for unbalanced data.
2. **Rely on `emmeans`**: Use **emmeans** to calculate unbiased marginal means, and `pairs()` to test the differences between them.

```
library(emmeans)

# Create a mock unbalanced dataset by dropping 3 observations
unbalanced_data <- fert_data[-c(1, 5, 9), ]
mod1 <- aov(yield ~ P * K, data = unbalanced_data)

# 1. Test main effects and interactions safely
joint_tests(mod1)
```

```

model term df1 df2 F.ratio p.value
P      1    9  11.117  0.0087
K      1    9   0.269  0.6164
P:K    1    9   0.371  0.5575

# 2. Get unbiased marginal means for Factor P
emm_P <- emmeans(mod1, ~ P)
emm_P

P      emmean   SE df lower.CL upper.CL
No_P   240 13.5  9      209      270
P      301 12.6  9      273      330

Results are averaged over the levels of: K
Confidence level used: 0.95

# 3. Test pairwise differences between those balanced means
pairs(emm_P)

contrast estimate    SE df t.ratio p.value
No_P - P      -61.6 18.5  9  -3.334  0.0087

Results are averaged over the levels of: K

```

9 Multiple Comparisons: Beyond the F-Test

Multiple Comparisons in Practice

- A significant ANOVA interaction tells you *some* difference exists, but testing all pairs blindly increases your *family-wise error rate* (false positive rate).
- The `emmeans` package provides a streamlined pipeline for post-hoc testing with appropriate data-dredging penalties.

```

# Assuming fert_mod is our fitted two-way model
emm_int <- emmeans(fert_mod, ~ P * K)

# Tukey adjustment for all pairwise comparisons
pairs(emm_int, adjust = "tukey")

```

Visualizing Contrasts

- Staring at massive tables of p-values makes it difficult to assess effect sizes.
- You can pass your contrast object directly into a plot to visually see which adjusted confidence intervals exclude zero.

```

# Generates a forest plot of the differences between groups
plot(pairs(emm_int, adjust = "tukey")) +
  ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  ggplot2::labs(title = "Tukey-Adjusted Pairwise Differences")

```

10 Resource Constraints: Unreplicated & Fractional Designs

Unreplicated Factorials ($n = 1$)

- Experiments are expensive. Sometimes we can only afford a single replicate per treatment combination ($n = 1$).
- *The Problem:* Fitting a full factorial model uses up all available degrees of freedom. The error degrees of freedom (df_E) becomes 0, so R returns NA for all standard errors and p-values.

```
# Fits main effects, 2-way, 3-way, and 4-way interactions
mod_unrep <- aov(yield ~ A * B * C * D, data = unrep_data)

# F-values and p-values will return as NA
summary(mod_unrep)
```

The Fix: Pooling into Error

- Relying on the *Sparsity of Effects Principle*, we assume high-order interactions (e.g., 3-way and 4-way) are effectively zero (just random noise).
- We restrict the model, pushing those negligible higher-order terms into the residual error to mathematically generate degrees of freedom.

```
# Restrict the model to main effects and 2-way interactions only
# The "~2" syntax drops 3-way and 4-way terms into the error
mod_pooled <- aov(yield ~ (A + B + C + D)^2, data = unrep_data)

# Now R can calculate p-values for main and 2-way effects!
summary(mod_pooled)
```

Fractional Factorials & Confounding

- A *fractional factorial* design runs only a structured fraction of the full combinations (e.g., 16 runs instead of 32 for a 5-factor design).
- **How R handles it:** R automatically recognizes the missing runs and returns NA for the interaction terms it cannot mathematically estimate.
- **Aliasing:** Because we skipped runs, certain effects are perfectly correlated (confounded).
 - *Intuition:* If we design a half-fraction by defining Factor C's levels mathematically as $A \times B$, R cannot tell if an observed change in yield is caused by the main effect of C, or the interaction of A and B.

```
# Generate a mock half-fraction dataset ( $2^{3-1}$ )
library(FrF2)
frac_data <- FrF2(nruns = 4, nfactors = 3, factor.names = c("A", "B", "C"))
frac_data$yield <- c(14, 18, 22, 19)

frac_mod <- aov(yield ~ A * B * C, data = frac_data)

# This outputs the exact matrix showing which main effects
# are mathematically identical to which interaction terms:
alias(frac_mod)
```

```

Model :
yield ~ A * B * C

Complete :
      (Intercept) A1 B1 C1
A1:B1      0      0  0  1
A1:C1      0      0  1  0
B1:C1      0      1  0  0
A1:B1:C1   1      0  0  0

```

Visual Diagnostics: Daniel Plots

- If you have zero degrees of freedom for error (saturated/fractional models), you can evaluate effects visually without an F-test using a *Half-Normal Probability Plot*.
- **Interpretation:** Noise falls along a straight line. Significant, active effects break away from the line.

```

library(FrF2)

# Visually identify active effects in an unreplicated or
# fractional design without needing an error term
DanielPlot(frac_mod, half = TRUE,
           main = "Half-Normal Plot of Effects")

```

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References

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5. [1](#)